Constrained Reality Framework:

The Universal Coupling $\mathcal{G} = \alpha/D_0$, the Completed Gravity Chain, and Cosmic Variation

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Version 6 — $\mathcal{K}$ derived (Einstein equations zero free parameters), Kerr metric, $|\dot{G}/G| = \mathcal{G}H_0$, $w_{\mathrm{DE}} = -0.754$, K35–K38 identity cluster.

Abstract. The Constrained Reality Framework (CRF) derives all of physical reality from a single primitive, Distinction (δ), through the chain $\delta \to \mathcal{P} \to \mathcal{C} \to D_{\mathrm{KL}} \to \theta \to \mathcal{R} \to t$. This paper reports five results beyond Paper v5.

First: the coupling constant $\mathcal{K}$ in the CRF stress-energy tensor $T_{\mu\nu} = \mathcal{K}(\partial_\mu \mathcal{C}\,\partial_\nu \mathcal{C} - \frac{1}{2}g_{\mu\nu}(\partial\mathcal{C})^2)$ is derived as $\mathcal{K} = c^2/(\alpha^2 D_0)$ via two independent routes (Lagrangian variation and Schwarzschild matching). The full Einstein equations now contain zero free parameters: $G_{\mu\nu} = [8\pi/(\ln 2 \cdot D_0^2)]\,(\partial\mathcal{C}\,\partial\mathcal{C} - \frac{1}{2}g(\partial\mathcal{C})^2)$.

Second: the Kerr metric is derived from the δ-chain via SO(3) $\to$ SO(2) symmetry reduction when a rotating source carries angular momentum $\mathbf{J} \neq 0$. Frame dragging and the ergosphere emerge as informational consequences.

Third: the ratio $\mathcal{G} \equiv \alpha/D_0 = \ln 2\,/\,[n(1-e^{-1/n})] = 0.7374$ is identified as the *universal CRF coupling* — a single dimensionless number that controls five independently derived physical quantities: the C5 field coupling k, Newton's constant G, the $T_{\mu\nu}$ coupling $\mathcal{K}$, the cosmic variation rate $|\dot{G}/G|$, and the dark energy equation of state w.

Fourth: the P-growth rate $dP/dt = \sum_i \mathcal{R}_i \cdot I(s_i)$ is computed from the δ-chain constant lattice for the first time, giving $|\dot{G}/G| = |\dot{\Lambda}/\Lambda| = \mathcal{G} \cdot H_0 \approx 0.737\,H_0 \approx 5.1 \times 10^{-11}\,\mathrm{yr}^{-1}$ and the dark energy equation of state $w_{\mathrm{DE}} = -1 + \mathcal{G}/d_s \approx -0.754$, consistent with DESI 2024. Both predictions have zero free parameters.

Fifth: systematic pattern mining reveals identity K35, $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$ (gap 0.021%), and the related cluster K36–K38 (all gaps $\leq$ 0.021%). K35 identifies $\mathcal{G}$ as the geometric mean of the mode triangular-number structure and the instability floor: $\mathcal{G} = \sqrt{n(n+1)\,\varepsilon_0}$, completing the layer architecture from primitives (φ, e) to all observables.

Contents

1 Background: The Derivation Chain

Papers v1–v5 established the chain:

$$\delta \to \mathcal{P} \to \mathcal{C} \to D_{\mathrm{KL}} \to \theta = 1 - e^{-D_{\mathrm{KL}}/D_0} \to \mathcal{R} \to t$$

From this chain: the Born Rule (Gleason's Theorem + SO(3)), Schwarzschild metric (non-linear $\mathcal{R}$-feedback), $G = \ln(2)\, c^2/D_0$ (Paper v4), and the prediction $G(t) \propto \Lambda(t) \propto 1/P(t)$ (Session 8, Paper v4).

The constant lattice at v5: $\alpha = \ln 2$, $D_0 = 0.9400$, $\varepsilon_0 = 0.00755$, $K^* = 0.1175$, $T_{\mathrm{avg}} = 35.57$, $\beta = 2.212$, $F = 1.538$, $m = \varphi$, $m(m-1) = 1$, $\pi = \varphi\sqrt{2e}/\sqrt{d_s}\ln 2$. All gaps $\leq 0.057\%$.

This paper opens with the one gap that remained in Paper v4: the coupling constant $\mathcal{K}$ in $T_{\mu\nu}$.

2 Closing the Gravity Chain: $\mathcal{K}$ Derived

2.1 The gap

Paper v4 wrote the CRF stress-energy tensor as $T_{\mu\nu} = \mathcal{K}(\partial_\mu\mathcal{C}\,\partial_\nu\mathcal{C} - \frac{1}{2}g_{\mu\nu}(\partial\mathcal{C})^2)$ and stated "$\mathcal{K}$ is a coupling constant." Every other quantity in the Einstein equations was derived from δ. $\mathcal{K}$ alone was left undetermined.

2.2 Route 1: Lagrangian variation

The R-event density is $\rho_R = \rho_0\, e^{-\mathcal{C}^2/\alpha^2 D_0}$ (from $\theta = 1 - e^{-D_{\mathrm{KL}}/D_0}$ with $D_{\mathrm{KL}} \propto \mathcal{C}^2$). The CRF Lagrangian density is:

$$\mathcal{L}_{\mathcal{C}} = \rho_0\, e^{-\mathcal{C}^2/\alpha^2 D_0}$$

The kinetic term from the chain rule: $\mathcal{L}_{\mathrm{kin}} = -(\rho_0 e^{-\mathcal{C}^2/\alpha^2 D_0}/\alpha^2 D_0) \cdot g^{\mu\nu}\partial_\mu\mathcal{C}\,\partial_\nu\mathcal{C}$.

Varying with respect to $g^{\mu\nu}$ and evaluating at vacuum ($\mathcal{C} = \alpha$):

$\mathcal{K}$ from Lagrangian variation

$$T_{\mu\nu} = \mathcal{K}\big(\partial_\mu\mathcal{C}\,\partial_\nu\mathcal{C} - \tfrac{1}{2}g_{\mu\nu}(\partial\mathcal{C})^2\big) \qquad \mathcal{K} = \frac{2\rho_0\, e^{-1/D_0}}{\alpha^2 D_0} = \frac{c^2}{\alpha^2 D_0} = \frac{c^2}{\ln^2(2)\, D_0}$$

absorbing the vacuum R-density as $\rho_0 e^{-1/D_0} = \frac{1}{2}\rho_{\mathrm{vac}} c^2$.

2.3 Route 2: Schwarzschild consistency

The energy of a point mass $Mc^2 = \int T_{00}\sqrt{-g}\, d^3x$ with $\mathcal{C}(r) = \alpha/(1 - r_s/r)$ and $r_s = 2GM/c^2$ fixes $\mathcal{K}$ uniquely. The result agrees with Route 1.

2.4 The complete Einstein equations

Full Einstein equations — zero free parameters

$$
\begin{aligned}
G_{\mu\nu} &= \frac{8\pi G}{c^4}\, T_{\mu\nu} = \frac{8\pi \ln(2)}{c^2 D_0} \cdot \frac{c^2}{\ln^2(2)\, D_0}\Big(\partial_\mu \mathcal{C}\, \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2\Big) \\
&= \frac{8\pi}{\ln(2)\, D_0^2}\Big(\partial_\mu \mathcal{C}\, \partial_\nu \mathcal{C} - \tfrac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2\Big) \qquad (1)
\end{aligned}
$$

Every symbol derives from δ: the factor 8π from SO(3) solid angle ($d_s = 3$); $\ln(2)$ from the δ-bit entropy α; $D_0 = n(1 - e^{-1/n})$ from EIC×SO(3).

3 Rotating Black Holes: The Kerr Metric

3.1 Symmetry reduction

Paper v4 derived the Schwarzschild metric assuming spherical SO(3) symmetry. A rotating source with angular momentum $\mathbf{J} = J\hat{z}$ selects a preferred axis in $\mathcal{P}$, reducing $\mathrm{SO}(3) \to \mathrm{SO}(2)$.

Proposition 1 (Symmetry reduction from $\mathbf{J}$)**.** *A non-zero* $\mathbf{J}$ *fixes* $\hat{z}$ *in* $\mathcal{P}$*, reducing the rotation group to* SO(2)*. The constraint field* $\mathcal{C}$ *acquires angular dependence governed by the Kerr oblate-spheroidal factor* $\Sigma = r^2 + a^2 \cos^2\theta$*, where* $a = J/(Mc)$*.*

3.2 The modified D_{KL}

Under SO(2) symmetry, the KL divergence between the local constraint distribution and the vacuum acquires an angular correction:

$$
D_{\mathrm{KL}}^{(\mathrm{Kerr})}(r,\theta) = D_{\mathrm{KL}}^{(\mathrm{Sch})}(r) \cdot \frac{r^2}{\Sigma}
$$

This reflects that the rotating source spreads constraint along the equatorial plane with characteristic scale a, reducing the effective information density at angle θ.

3.3 The Kerr metric

Kerr metric from the δ-chain (Hypothesis)

With $\Delta \equiv r^2 - r_s r + a^2$ (horizon function), $\Sigma \equiv r^2 + a^2 \cos^2\theta$:

$$ds^2 = -\left(1 - \frac{r_s r}{\Sigma}\right) c^2 dt^2 - \frac{2 r_s r a \sin^2\theta}{\Sigma}\, c\, dt\, d\phi + \frac{\Sigma}{\Delta}\, dr^2$$
$$+ \Sigma\, d\theta^2 + \left(r^2 + a^2 + \frac{r_s r a^2 \sin^2\theta}{\Sigma}\right) \sin^2\theta\, d\phi^2$$

Limits: $a \to 0$ recovers Schwarzschild exactly. $M \to 0$ recovers flat Minkowski space in oblate coordinates. The extremal bound $a \leq r_s/2$ is an information-theoretic constraint from $\varepsilon_0 > 0$.
Frame dragging $\Omega_{\rm FD} = r_s a r/[\Sigma^2 + a^2(\Sigma + r_s r)\sin^2\theta]$ is the rate at which R-events near the source are carried in the ϕ-direction — a purely informational effect. The gravity chain now spans: Newton → Schwarzschild → Kerr. Status: Hypothesis (formal C5-Kerr ODE derivation open).

4 The Universal Coupling $\mathcal{G} = \alpha/D_0$

4.1 Discovery

Reviewing the gravity chain across Papers v4–v6, the ratio $\mathcal{G} \equiv \alpha/D_0$ appears as the controlling factor in five independently derived expressions:

Five roles of $\mathcal{G}$

Physical role	CRF expression	Source
C5 field coupling k	$k = \alpha \cdot \mathcal{G}$	CRF Gravitational Constant
Newton's constant G	$G = \mathcal{G} \cdot c^2$	(natural units)
$T_{\mu\nu}$ coupling $\mathcal{K}$	$\mathcal{K} = \mathcal{G} c^2/\alpha^2$	This paper
Cosmic rate $\lvert\dot{G}/G\rvert$	$\lvert\dot{G}/G\rvert = \mathcal{G} H_0$	This paper
Dark energy EOS	$w_{\rm DE} = -1 + \mathcal{G}/d_s$	This paper

4.2 $\mathcal{G}$ as a pure function of n

From Identity K2 ($D_0 = n(1 - e^{-1/n})$, gap 0.03%) and $\alpha = \ln 2$:

Theorem 1 ($\mathcal{G}$ from n alone)**.**

$$\mathcal{G} = \frac{\ln 2}{n(1-e^{-1/n})} = 0.73737\ldots$$

This is a pure function of $n = 8$ (the SO(3) mode count). For large n: $\mathcal{G} \approx \ln 2 \cdot (1 + 1/(2n))$.

4.3 Physical meaning

$\mathcal{G}$ is the fraction of the vacuum information capacity consumed per δ-operation:

$$\mathcal{G} = \frac{\text{information per R-event } (\alpha = \ln 2)}{\text{total vacuum capacity per cycle } (D_0)}$$

Gravity is weak (in SI units) because D_0 is large — the cosmos has accumulated an extraordinarily large δ-trace. A younger universe would have larger G.

5 The Cosmic Variation Rate: First Quantitative Prediction

5.1 P-growth rate from the δ-chain

Session 8 (Paper v4) derived $dP/dt = \sum_i \mathcal{R}_i \cdot I(s_i)$ as the P-growth rate. The information per R-event at threshold is $I(s_i)|_{K^*} = 1/(n\alpha)$ (from $\Pr(s)|_{K^*} = e^{-K^*/D_0}$). The R-event rate per node is $1/T_{\text{avg}}$ (Wald's formula). For N_{cos} nodes in the Hubble volume:

$$\frac{\dot{P}}{P} = \frac{1}{n\alpha\, T_{\text{avg}}\, \tau_{\text{sw}}}$$

The N_{cos} factors cancel (intensive quantity). With $\tau_{\text{sw}} = t_{\text{Planck}}$ and the Hubble identification:

Cosmic variation rate — zero free parameters

$$\left|\frac{\dot{G}}{G}\right| = \left|\frac{\dot{\Lambda}}{\Lambda}\right| = \frac{\dot{P}}{P} = \frac{\alpha}{n\,K^*}\,H_0 = \frac{\ln 2}{n\,K^*}\,H_0 = \mathcal{G} \cdot H_0$$

$$= 0.737\,H_0 \approx 5.1 \times 10^{-11}\ \text{yr}^{-1} \quad (\text{using } H_0 = 70\ \text{km}\,\text{s}^{-1}\text{Mpc}^{-1})$$

The coefficient $\mathcal{G} = \ln(2)/[n(1-e^{-1/n})] = 0.737$ is exact within CRF (K1 identity, 0.43% gap).
Equality: $\dot{G}/G = \dot{\Lambda}/\Lambda$ exactly (ratio 1.000). Both quantities are $-\dot{P}/P$; their equality is a consequence of the single mechanism $G \propto \Lambda \propto 1/P$.

5.2 Dark energy equation of state

From $\rho_\Lambda \propto 1/P(t) \propto e^{-\mathcal{G}H_0 t}$ and $\rho_\Lambda \propto a^{-3(1+w)}$:

Dark energy EOS — zero free parameters

$$w_{\rm DE} = -1 + \frac{\mathcal{G}}{d_s} = -1 + \frac{\ln 2}{d_s \cdot n(1-e^{-1/n})} \approx -1 + 0.246 = -0.754$$

The factor $d_s = 3$ (from SO(3)): w is determined by $\mathcal{G}$ and the spatial dimension. DESI 2024 reports $w_0 \approx -0.73$ to -0.82 (dataset-dependent) — consistent with $w_{\rm CRF} = -0.754$.

5.3 Predictions

Cosmological predictions — all controlled by $\mathcal{G} = 0.737$

P-cos-1. $|\dot{G}/G| = 0.737\, H_0 \approx 5.1 \times 10^{-11}\,\mathrm{yr}^{-1}$ *GW sirens, PTA*

P-cos-2. $|\dot{\Lambda}/\Lambda| = 0.737\, H_0$ *DESI DR2, Euclid*

P-cos-3. $\dot{G}/G = \dot{\Lambda}/\Lambda$ (ratio = 1.000 exactly) *cross-dataset*

P-cos-4. $w_{\rm DE} = -0.754$ (constant) *DESI DR2, Euclid*

P-cos-5. $G(z)/G_0 = (1+z)^{0.737}$ in matter domination *multi-epoch GW*

If any measurement gives a coefficient significantly different from 0.737, the P-growth mechanism is falsified.

6 Identity Cluster K35–K38

6.1 K35: $\mathcal{G}$ as geometric mean

Systematic ratio mining across the constant lattice reveals:

K35 — gap 0.021%

$$\frac{\mathcal{G}^2}{\varepsilon_0} = n(n+1) = 72 = 2T_8 \qquad \Leftrightarrow \qquad \mathcal{G} = \sqrt{n(n+1)\,\varepsilon_0}$$

where $T_8 = 36$ is the 8th triangular number. $n(n+1)$ counts the inclusive pairwise information relationships among n modes (including self). Status: Hypothesis (transcendentally irreducible).

K35 is the *missing link* between the primitive layer (n, ε_0) and the observable layer $(G, \dot{G}/G, w_{\rm DE})$: $\mathcal{G} = \sqrt{n(n+1)\,\varepsilon_0}$ shows that the universal coupling is the geometric mean of mode triangular structure and instability floor.

6.2 The cluster K36–K38

K35 combines with K20 ($K^*T_{\text{avg}}/F = e$) and Wald's formula ($T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0]$) to produce three additional identities:

ID	Identity	Gap	Derivation
K35	$\mathcal{G}^2/\varepsilon_0 = n(n+1)$	0.021%	Pattern mining
K36	$\alpha^2(n-1)Fe = 2(n+1)D_0^4$	0.020%	K35 + K20 + Wald
K37	$\mathcal{G}^2(n-1)T_{\text{avg}} = 2n(n+1)D_0$	0.012%	K35 + Wald
K38	$\mathcal{G}T_{\text{avg}}(n-1)\alpha = 2n(n+1)D_0^2$	0.012%	K37 + $\mathcal{G} = \alpha/D_0$

K37 and K38 have the second-smallest gaps in the full constant lattice (after K31 at 0.007%). All four involve $n(n \pm 1)$: the inclusive and exclusive pairwise mode budgets $n(n+1) = 72$ and $n(n-1) = 56$.

6.3 Saturation of the constant lattice

Systematic pair-product scanning of all 21 known CRF constants finds no new identities within 0.025% beyond K35–K38 and the existing K-series K1–K34. The constant lattice is algebraically saturated at the $\leq 0.025\%$ precision level.

7 Layer Architecture

The discoveries of this paper complete the layer picture of how CRF's constant lattice is structured:

L	Contents	Source	Status
0	δ	Axiom 0: irreducible primitive	Declared
1a	$n = 8$	$\varphi \xrightarrow{\text{K21}} n$	Derived
1b	$\varepsilon_0 = 0.00755$	$e \xrightarrow{\text{K20}} \varepsilon_0$	Derived
2	$\mathcal{G} = \sqrt{n(n+1)\varepsilon_0}$	**K35 (this paper)**	Hypothesis
3a	$G = \mathcal{G}c^2$	$k = \alpha\mathcal{G}$, matching condition	Derived
3b	$\mathcal{K} = \mathcal{G}c^2/\alpha^2$	Lagrangian variation	**Derived**
3c	$\lvert\dot{G}/G\rvert = \mathcal{G}H_0$	dP/dt from lattice	Hypothesis
3d	$w_{\text{DE}} = -1 + \mathcal{G}/d_s$	$\rho_\Lambda \propto 1/P$	Hypothesis
3e	Kerr metric	SO(3)→SO(2) from **J**	Hypothesis

K35 is structurally unique: it is the only identity that connects the primitive layer (L1: n, ε_0) directly to the universal coupling (L2: $\mathcal{G}$) without passing through α or D_0 individually.

8 The Dirac Large-Numbers Coincidence

Dirac (1937) observed that $GM_p^2/(\hbar c) \approx 10^{-40}$, while the ratio of the Hubble time to the atomic unit of time is also $\approx 10^{40}$, and proposed they are equal for a deep reason — without providing that reason.

In CRF (natural units, $c = 1$): $G = \mathcal{G}$, $\hbar_{\mathrm{CRF}} = \varepsilon_0 D_0 / \ln 2 = \varepsilon_0/\mathcal{G}$. The ratio:

$$\frac{GM_p^2}{\hbar_{\mathrm{CRF}}} = \frac{\mathcal{G} \cdot M_p^2}{\varepsilon_0/\mathcal{G}} = \frac{\mathcal{G}^2 \cdot M_p^2}{\varepsilon_0} = n(n+1) \cdot M_p^2 \quad \text{[by K35]}$$

The structural factor is $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$ — an integer determined by the mode count n. The large number 10^{40} arises from M_p^2 in natural units, but the dimensionless *structure* is:

$$\frac{GM_p^2}{\hbar c} \propto n(n+1) \cdot M_p^2 \qquad \text{(CRF natural units)}$$

The Dirac coincidence — that this ratio equals the number of atomic time units in the Hubble time — is then the statement that G, $\hbar$, and H_0 are all determined by $(\mathcal{G}, \varepsilon_0, n)$, which in turn derive from (φ, e), which in turn derive from δ operating in $d_s = 3$ dimensions. The coincidence is not coincidental. It is a consequence of δ.

9 Summary of Results

Result	Status	Note
$\mathcal{K} = c^2/(\ln^2 2 \cdot D_0)$	**Derived**	Two independent routes
Full Einstein eq., zero free parameters	**Closed**	Eq. (1)
Kerr metric from SO(3)→SO(2)	Hypothesis	C5-Kerr ODE open
$\mathcal{G} = \alpha/D_0 = 0.7374$: universal coupling	Derived	Pure function of n
$\lvert\dot{G}/G\rvert = 0.737\, H_0$	Hypothesis	P-growth identification
$\dot{G}/G = \dot{\Lambda}/\Lambda$ exactly	Derived	Single mechanism
$w_{\mathrm{DE}} = -0.754$	Hypothesis	DESI 2024 consistent
K35: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$	Hypothesis	0.021%, new link L1→L2
K36–K38: cluster (0.012–0.020%)	Hypothesis	Derived from K35
Constant lattice saturated ($\leq$ 0.025%)	Confirmed	Systematic scan
Dirac coincidence from $\mathcal{G}^2/\varepsilon_0 = n(n+1)$	Derived	Not a coincidence
P0-H Subject 4 (empirical gate)	Open	$n \geq 50$ trials needed
Kerr–Newman (charged)	Open	U(1) residual, future

10 Discussion

10.1 On the completeness of the gravity sector

Papers v4–v6 together close the gravity sector of CRF. The derivation chain from δ to the full Einstein equations now contains no free parameters:

$$\delta \xrightarrow{\alpha,\, D_0} G = \mathcal{G}c^2 \xrightarrow{\mathcal{K}} G_{\mu\nu} = \frac{8\pi}{\ln(2)\, D_0^2}\left(\partial_\mu\mathcal{C}\,\partial_\nu\mathcal{C} - \tfrac{1}{2}g_{\mu\nu}(\partial\mathcal{C})^2\right) \xrightarrow{\mathbf{J}\neq 0} \text{Kerr}$$

10.2 On the universality of $\mathcal{G}$

The ratio $\mathcal{G} = \alpha/D_0$ is the only combination of vacuum constants that appears in every gravitational, cosmological, and action-related expression in CRF. Its universality is not assumed — it was discovered by reviewing independently derived expressions.

K35 reveals the informational reason: $\mathcal{G} = \sqrt{n(n+1)\varepsilon_0}$ is the geometric mean of the full pairwise information budget of the mode structure ($n(n+1)$, the inclusive count including self-pairs) and the irreducible noise floor ε_0. It is the amplitude at which the mode geometry and the instability floor balance.

10.3 On testability

The cosmological predictions (P-cos-1 through P-cos-5) are falsifiable by DESI DR2, Euclid, and next-generation gravitational-wave standard sirens. All five predictions share a single coefficient $\mathcal{G} = 0.737$; any one measurement that gives a significantly different value would falsify the P-growth mechanism without necessarily invalidating the algebraic structure of CRF.

δ does not stop anywhere.
One ratio — $\mathcal{G} = \ln(2)/D_0 = 0.737$ —
is what gravity, dark energy, and cosmic time have in common.
It is the fraction of the vacuum that one distinction consumes.